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Does Gassmann Fluid Substitution Work in Complex Carbonates?

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Obtaining or monitoring fluid saturation from elastic properties (such as quantitative seismic interpretation) is of great importance to oil and gas industry as well as gas storage applications (Avseth et al. 2000; Fawad et al., 2020; Dvorkin et al. 2014; Fawad and Mondol, 2022). The most commonly used formulations for linking fluid content/saturation and elastic properties is Gassmann (Gassmann 1951). Gassmann theory has been used extensively in rock-physics related literature to either test the feasibility of time lapse seismic (forward modeling) or to invert for changes in fluid saturation based on time lapse seismic (Avseth et al. 2000; Mavko et al., 2009; Dvorkin et al. 2014; Fawad et al., 2020; Fawad and Mondol, 2022). Such applications have gained popularity especially in sandstone reservoirs while the applicability of Gassmann in carbonates have been a debate for over a decade.

Carbonate rocks are frequently characterized by heterogeneous microstructure and properties as complicated by their variable pore types and geometry (e.g., Eberli et al., 2003; Fournier et al., 2014; 2018; Neto et al., 2014; El Husseiny and Vanorio, 2015; 2017; El-Husseiny et al., 2019; 2022; Regnet et al., 2015; Jaballah et al., 2021; Salih et al., 2021, Reijmer et al., 2021). Such heterogeneity combined with the reactivity of calcite and dolomite mineralogy have raised questions about whether Gassmann theory can be used in carbonates or not (Adam et al., 2006; Vega et al., 2010; Sharma et al., 2013; Vanorio et al., 2011). Large body of literature, including mainly laboratory measurements at ultrasonic frequencies indicated that Gassmann formulations tend to underestimate the changes in velocities associated with saturation changes (Baechle et al., 2005; Rogen et al., 2005; Khodja et al., 2022). Nevertheless, an overestimation by Gassmann's prediction has been observed for some cases/samples (Wang, 2000; Baechle et al., 2005; Vega et al., 2010; El-Husseiny et al., 2019).

Due to the scarcity of well log data and the feasibility of lab measurements, most studies conducted to evaluate the applicability of Gassmann have utilized the laboratory ultrasonic measurements and core samples (Vega et al., 2010; Sharma et al., 2013; Baechle et al., 2005; Vanorio et al., 2011; Khodja et al., 2022). The conventional approach is to measure elastic properties before (i.e., dry) and after saturation with a fluid and then compare Gassmann predictions with the measured saturated values. One fundamental issue with this approach is that ultrasonic measurements of saturated rock samples are prone for dispersion (i.e., dependent of velocity on frequency) which is not present at low frequency (at which Gassmann is applicable). Many studies have discussed this dispersion and its impact on velocity (e.g., Mavko and Jizba, 1991; Borgomano et al., 2017) which is not applicable to seismic frequency and is not considered

in Gassmann. Another fundamental issue with many studies is the use of a fluid that is not at equilibrium with the rock (like distilled or deionized water) which then results in dissolution for example and shear weakening (Adam et al., 2006; Baechle et al., 2005; Vanorio et al., 2011). Such weakening clearly would violate Gassmann condition for constant shear modulus during the fluid substitution. In avoiding the above issues, this work investigates the applicability of Gassmann in carbonates using low frequency laboratory measurements published in literature in addition to combining ultrasonic laboratory measurements from dry cores (should not suffer from dispersion) with well log data. In particular, this study aims to answer the following questions:

1. Does Gassmann provide accurate prediction in carbonates when dealing with fluids that are not reactive (i.e., like oil or brine in equilibrium with the rock frame)?
2. Can Gassmann be used in rock-physics modeling of carbonates to build the velocity-porosity relationship considering different fluid content?

Introduction

Obtaining or monitoring fluid saturation from elastic properties (such as quantitative seismic interpretation) is of great importance to oil and gas industry as well as gas storage applications (Avseth et al. 2000; Fawad et al., 2020; Dvorkin et al. 2014; Fawad and Mondol, 2022). The most commonly used formulations for linking fluid content/saturation and elastic properties is Gassmann (Gassmann 1951). Gassmann theory has been used extensively in rock-physics related literature to either test the feasibility of time lapse seismic (forward modeling) or to invert for changes in fluid saturation based on time lapse seismic (Avseth et al. 2000; Mavko et al., 2009; Dvorkin et al. 2014; Fawad et al., 2020; Fawad and Mondol, 2022). Such applications have gained popularity especially in sandstone reservoirs while the applicability of Gassmann in carbonates have been a debate for over a decade.

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issues, this work investigates the applicability of Gassmann in carbonates using low frequency laboratory measurements published in literature in addition to combining ultrasonic laboratory measurements from dry cores (should not suffer from dispersion) with well log data. In particular, this study aims to answer the following questions:

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2. Can Gassmann be used in rock-physics modeling of carbonates to build the velocity-porosity relationship considering different fluid content?

Theoretical Background

Gassmann's formulation is widely used to predict the effective bulk modulus of any saturated rock (K_{sat}) based on bulk modulus of the dry rock (K_{dry}) using the equation below (Gassmann 1951):

$$K_{sat} = K_{dry} + \frac{(1 - K_{dry}/K_o)^2}{\phi/K_{fl} + (1 - \phi)/K_o - K_{dry}/K_o^2} \quad (1)$$

In equation 1, ϕ is the porosity (in fraction), K_o is the bulk modulus of the pure mineral, while K_{dry} is obtained from the measured V_p , V_s and bulk density (ρ) of the dry case as per the following the equation:

$$K_{dry} = \rho \left(V_p^2 - \frac{4}{3} V_s^2 \right) \quad (2)$$

The effective fluid bulk modulus, K_{fl} , is obtained using Batzle and Wang (1992) equation for pure fluids (e.g., K_{brine} and K_{oil}) while for a mixture of fluids, the following saturation-weighted harmonic average can be used:

$$\frac{1}{K_{fl}} = \frac{s_w}{K_{brine}} + \frac{1 - s_w}{K_{oil}} \quad (3)$$

Finally, the K_{sat} value obtained from Eq. 1 is used to calculate the compressional as well as the shear wave velocities at the saturation condition (V_{p2} , V_{s2} respectively):

$$V_{p2} = \sqrt{\left(K_{sat} + \frac{4}{3} \mu_{sat} \right) / \rho_{sat}} \quad (4)$$

$$V_{s2} = \sqrt{\left(\mu_{sat} \right) / \rho_{sat}} \quad (5)$$

$$\rho_2 = \rho_{calcite} (1 - \phi) + \phi \rho_{oil} (1 - S_w) + \phi \rho_{brine} (S_w) \quad (6)$$

In equations 4 and 5, μ_{sat} is the saturated rock shear modulus which is equal to shear modulus at dry condition as per Gassmann's theory in which shear modulus is constant (i.e., fluid substitution does not result in frame changes). It is very evident here that Gassmann theory should not be expected to accurately predict the impact of fluid changes on elastic response when the fluids cause changes in shear modulus. Such changes can be for example associated with dissolution or precipitation due to fluid changes (such as CO₂ injection in carbonates; Vanorio et al., 2011). Another important consideration in Gassmann is the low frequency condition in which fluids will have enough time flow eliminating the wave-induced pressure gradient (Mavko and Jizba, 1991).

Materials and Method

We used subset from the core data (65 core plugs) utilized by El-Husseiny et al. (2022) whose main focus was to investigate the scalability issue between core data and well logs. We focus rather here in evaluating

Gassmann applicability in carbonates and by also adding data from literature including low frequency laboratory measurements of velocity saturated with brine and butane (Adam et al., 2006).

The first dataset consists of 65 core plugs retrieved from 3 wells in the middle east. The samples are all limestones (calcite mineralogy) with texture that varies from wackstone to grainestone. The porosity of all samples was measured using helium porosimeter while the compressional and shear wave velocity (V_p and V_s respectively) were measured under confining pressure of 35 MPa (similar to subsurface reservoir differential pressure) at dry conditions. Gassmann fluid substitution formulations were then used to calculate V_p for the full brine saturation where density and bulk modulus of brine were 1 g/cc and 2.63 GPa respectively (as obtained from Batzle and Wang, 1992). The use of dry measurements as inputs in the Gassmann ensures that velocity dispersion is not an issue since dry porous media do not typically show frequency-dependent behavior. The predictions from Gassmann based on core laboratory measurements are then compared to the V_p values obtained from well logs (already at brine bearing situation). To overcome the scalability (upscaling) and representation issues, we will be rather comparing the velocity-porosity relationships obtained from cores versus these from well logs. Recent studies show that the use of "trends" or relationship between physical quantities can be better used for upscaling or when comparing measurements done at different scales, rather than comparing absolute values at particular depths (Dvorkin and Nur, 2009; El-Husseiny et al., 2022). Consistency between the velocity-porosity relationships obtained from core data (after Gassmann fluid substitution) and the in-situ well logging data at the same fluid saturation would suggest the validity of Gassmann.

For the second dataset, it consists of 8 carbonate samples that include both limestone and dolostone extracted from a single subsurface reservoir well (Adam et al., 2006). Measurements were conducted at two different frequency ranges and at two different differential pressures (3.5 MPa and 31 MPa). The low seismic frequency moduli and velocity were obtained using the stress-strain approach as detailed by Batzle et al., (2006) while ultrasonic measurements were used for comparison to investigate the impact of frequency. The measurements were conducted on dry samples and then repeated after saturation with two different fluids: brine and Butane. The latter fluid should not cause much interaction with the solid rock frame. Dry measurements are again used as inputs for Gassmann theory to predict the moduli at the full brine and then full Butane saturations which are then compared with laboratory measurements conducted on the saturated samples.

Result and Discussion

Figure 1 shows the V_p -porosity relationship obtained based on well logs (reference data) and based on dry core data after applying Gassmann fluid substitution. The V_p -porosity relationship that is obtained from core data is consistent with those obtained from well logs. This consistency indicate that the use of Gassmann is valid in carbonates for building or modifying rock-physics transforms. In practice, the main significance of Gassmann theory is in building rock-physics transforms that can related elastic response with rock properties which is validated here. While the trend is accurately captured as shown in Figure 1, the velocity magnitude (at any given porosity) is also predicted well to the first order.

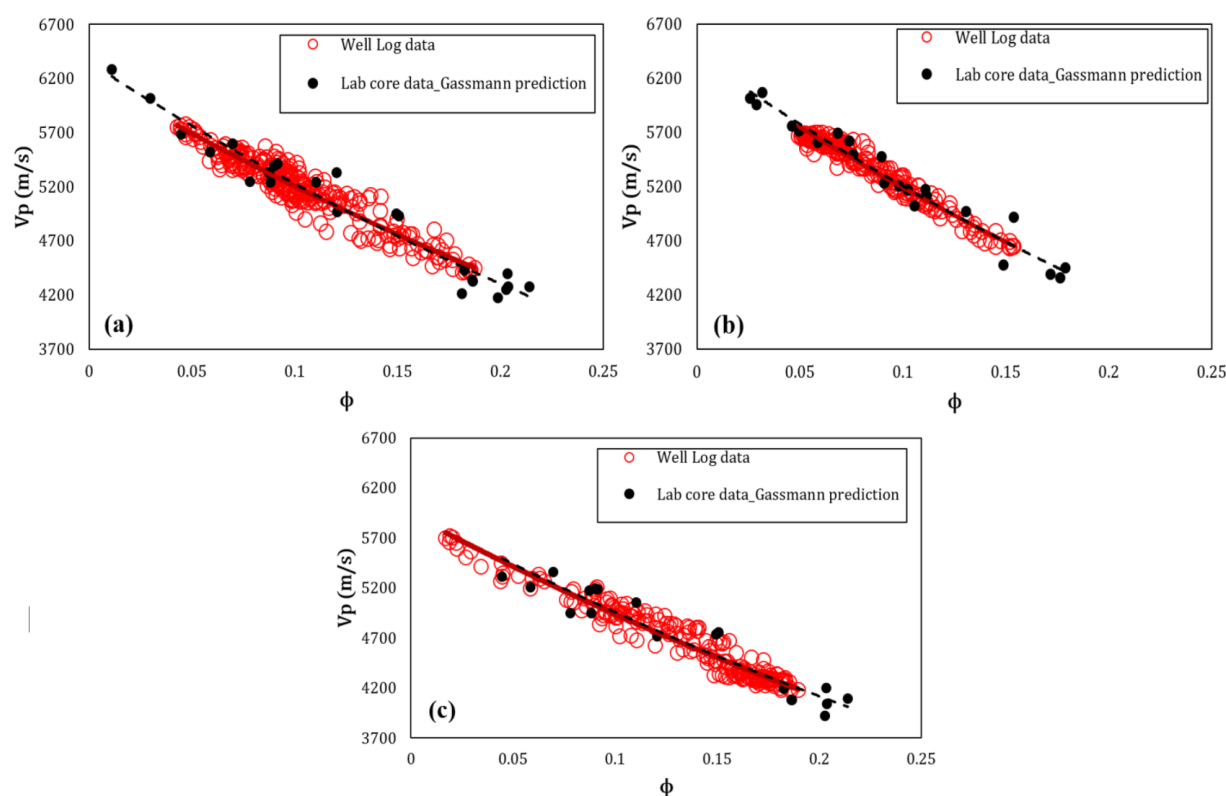


Figure 1—Vp-porosity relationship comparing data from well log data (red) with that from core plugs (black). The Vp of cores was obtained by using the dry Vp as input in Gassmann fluid substitution to obtain Vp at saturation. The velocity-porosity observed from each data are showing consistent trends.

The data from Adam et al., (2006) can be examined at different steps. First, let us examine the shear modulus results at low frequency as shown in Figure 2 for both brine and Butane. Ideally, there should be no change in shear modulus between dry and saturated case since shear modulus is not sensitive to fluid changes. It is evident that brine saturated samples experienced shear weakening as the shear modulus dropped a bit compared to the dry case (Figure 2). In contrast, saturation with butane resulted in negligible changes in shear modulus, significantly smaller than the case of brine. Shear weakening might be explained by reduction in surface energy or micro cracks growth due to interaction between high surface area pores (i.e., compliant pores) and polar fluids like brine (Adam et al., 2006; Tutuncu and Sharma, 1992). Such interaction is highly reduced in non-polar fluids like butane. The shear weakening observed in Figure 2 for the brine case has been reflected in the applicability of Gassmann which does not consider shear weakening. As evident from Figure 3, Gassmann in the case of brine tends to overestimate the bulk modulus of the saturated rocks. On the other hand, the bulk modulus of Butane saturated rocks was predicted reasonably well with Gassmann theory. The main difference between brine and butane can be the non-reactive nature of the butane while brine causes shear weakening as it has polar nature and it is not in chemical equilibrium with the rock (it is not a formation brine).

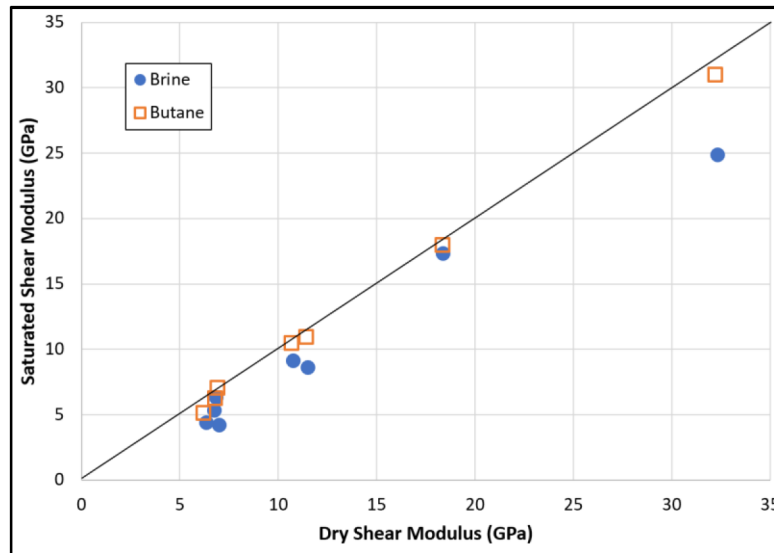


Figure 2—Comparison between the shear modulus at dry and saturated case for brine and butane fluids for low frequency measurements (100 Hz) for data obtained by Adam et al. (2006). Saturated shear modulus shows shear weakening (lower values) compared to dry shear modulus while Butane saturated rocks showed better match.

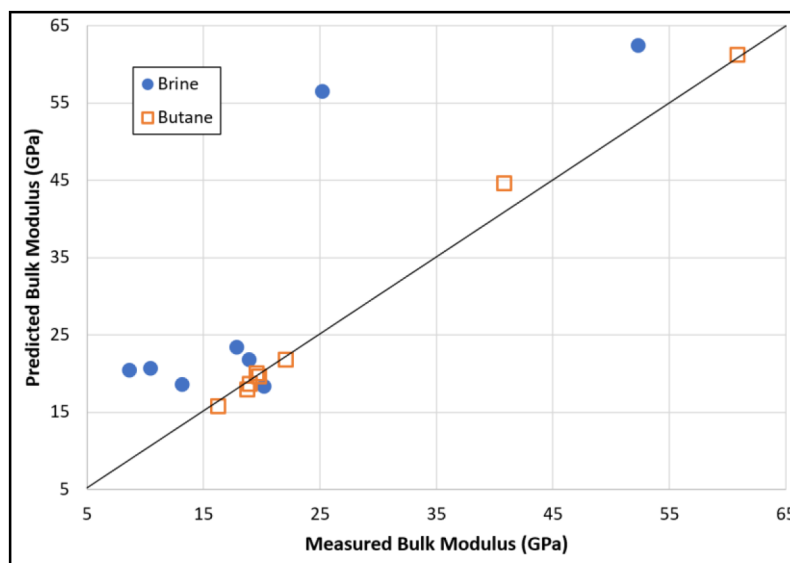


Figure 3—Comparison between the measured bulk modulus and predicted one (based on Gassmann) at saturated case for brine and butane fluids for low frequency measurements (100 Hz) where data was obtained by Adam et al. (2006). Gassmann theory predicted very well the data for butane saturated samples but overestimated the bulk modulus for the brine case.

The results in Figs. 1–3 indicate that Gassmann theory does work for carbonates as long as the fluids (either original or new) are in equilibrium with the rock frame and as long as low frequency condition is met. To confirm this, we include here a summary of major studies that reported underestimation or overestimation by Gassmann theory in Carbonates, along with the fluid and frequency used. Table 1 summarizes major studies that examined the applicability of Gassmann utilizing ultrasonic measurements. One major take away is that the conclusion of the studies in this case depends heavily in the fluid used. In case of distilled water, carbonated water (CO₂-brine) or brine (polar or reactive fluid), Gassmann theory tends to overestimate the changes in velocity due to shear weakening as the fluid interacts with the solid frame. On the other hand, the use of none polar fluids such as oil can result in stiffening at the ultrasonic frequency (dispersion) which is underpredicted by Gassmann. The effect of shear weakening is sometime overtaken (or balanced) by the dispersion (shear stiffening) resulting in overall apparent match between

Gassmann predictions and measurements. That is, it is unlikely valid to evaluate Gassmann fluid substitution theory in Carbonates using fluids that are not in equilibrium with the rock or using ultrasonic measurements. For example, Gassmann fluid substitution should not be used alone for CO₂-related geo-storage projects. Replacement of brine by CO₂ is expected to decrease velocity but the actual velocity after CO₂ injection is likely to be smaller than Gassmann prediction due to the expected impact of dissolution for example. In this case, predicting the change in velocity might be done at two steps where Gassmann theory can predict the changes due to replacing brine (higher bulk modulus fluid) by CO₂ or carbonated water, and then experimental data can inform the additional expected softening due to dissolution. Nevertheless, it should be valid to use Gassmann theory to predict changes in elastic properties due to production for example in gas or oil reservoirs since the formation brine and hydrocarbons are in chemical equilibrium with the rocks (i.e., original fluids in the system).

Table 1—Summary of various studies that examined the performance of Gassmann fluid substitution for carbonates by using ultrasonic measurements.

Study	Fluid used	Frequency	Remarks
Assefa et al., (2003)	Distilled water	Ultrasonic (0.85 MHz)	Shear weakening observed resulting in overestimation of velocity by Gassmann
Baechle et al., (2005)	Distilled water	Ultrasonic (1 MHz)	Both overestimation and underestimation were observed. Gassmann underpredict the velocity when there is velocity dispersion (causing shear stiffening) while Gassmann's overpredictions are associated with cases where shear weakening overcome the impact of dispersion.
Vanorio et al. (2010)	CO ₂ -brine & distilled water	Ultrasonic (0.75-1 MHz)	Gassmann overpredicts the P-wave velocity for the CO ₂ -brine saturated rocks while mainly underpredicts the values for distilled water
Khodja et al., (2021)	Brine and crude oil	Ultrasonic (0.75-1 MHz)	Samples saturated with oil showed stiffening that was not considered in Gassmann theory (underprediction from Gassmann)

Conclusions

This study examined whether Gassmann fluid substitution is applicable for use in carbonates by utilizing experimental core data, well logs and published literature. The main outcomes indicate that Gassmann theory should be applicable in carbonates as long as the fluids are in equilibrium with the host rock matrix to insure no violation for the constant shear modulus condition. This also extend to the use for establishing rock physics transforms such as the velocity-porosity relationship. One additional condition is the low frequency which was not met by most of earlier studies resulting in non-consistent observations that are mainly due to complications by dispersion. The latter has resulted in over or under estimation by Gassmann depending on the interplay with shear weakening. In summary, it should be valid to use Gassmann theory to predict changes in elastic properties (at seismic or well log frequency) due to production for example in gas or oil reservoirs since the formation brine and hydrocarbons are not reactive with respect to the host rock.

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